

Saudi Retail Business Intelligence Dashboard Report

Power BI report for retail sales performance, customer behavior, inventory risk, and branch operations in Saudi Arabia.

Prepared by	Yasir Awad - Data Analyst
Project Type	Freelance Retail BI / Power BI Dashboard
Domain	Retail Sales Customer Analytics Inventory Operations
Tools	Power BI, DAX, Power Query, Data Modeling, Excel/CSV
Report Purpose	Document dashboard objectives, business questions, insights, and recommendations.

965.79K	12K	375.35K	38.86%
Total Revenue	Total Orders	Total Profit	Profit Margin

Confidentiality note: Client/store identity and sensitive operational details are anonymized. Product names are generic where necessary.

1. Executive Summary

This report documents a Power BI Business Intelligence dashboard developed for a retail business operating in Saudi Arabia. The dashboard centralizes sales, customer, inventory, and branch performance metrics into a structured reporting solution for management decision-making.

The project focused on converting raw retail operational data into executive-level dashboards that help business owners monitor performance, identify high-value customers, detect inventory risk, and evaluate branch operations.

Business Problem

The business lacked a centralized view of key operational and sales metrics. Management needed a clear way to monitor sales trends, customer behavior, inventory risks, and branch-level performance in one place.

Project Objectives

- Build an executive overview for high-level KPI monitoring.
- Analyze revenue trends, product contribution, and branch performance.
- Understand customer behavior, returning customer contribution, and customer value segmentation.
- Monitor low-stock items, reorder requirements, and inventory risk by branch.
- Provide actionable recommendations to support sales, inventory, and operational decisions.

Dashboard Structure

Dashboard Page	Business Purpose
Executive Overview	High-level KPI reporting and summary metrics for management.
Sales Performance Analysis	Revenue trends, product performance, category contribution, and branch comparison.
Customer Behavior Analysis	Customer segmentation, returning vs new customer contribution, and retention indicators.
Inventory & Operations Analysis	Inventory health, low-stock products, reorder signals, and operational inventory risk.

2. Business Questions Addressed

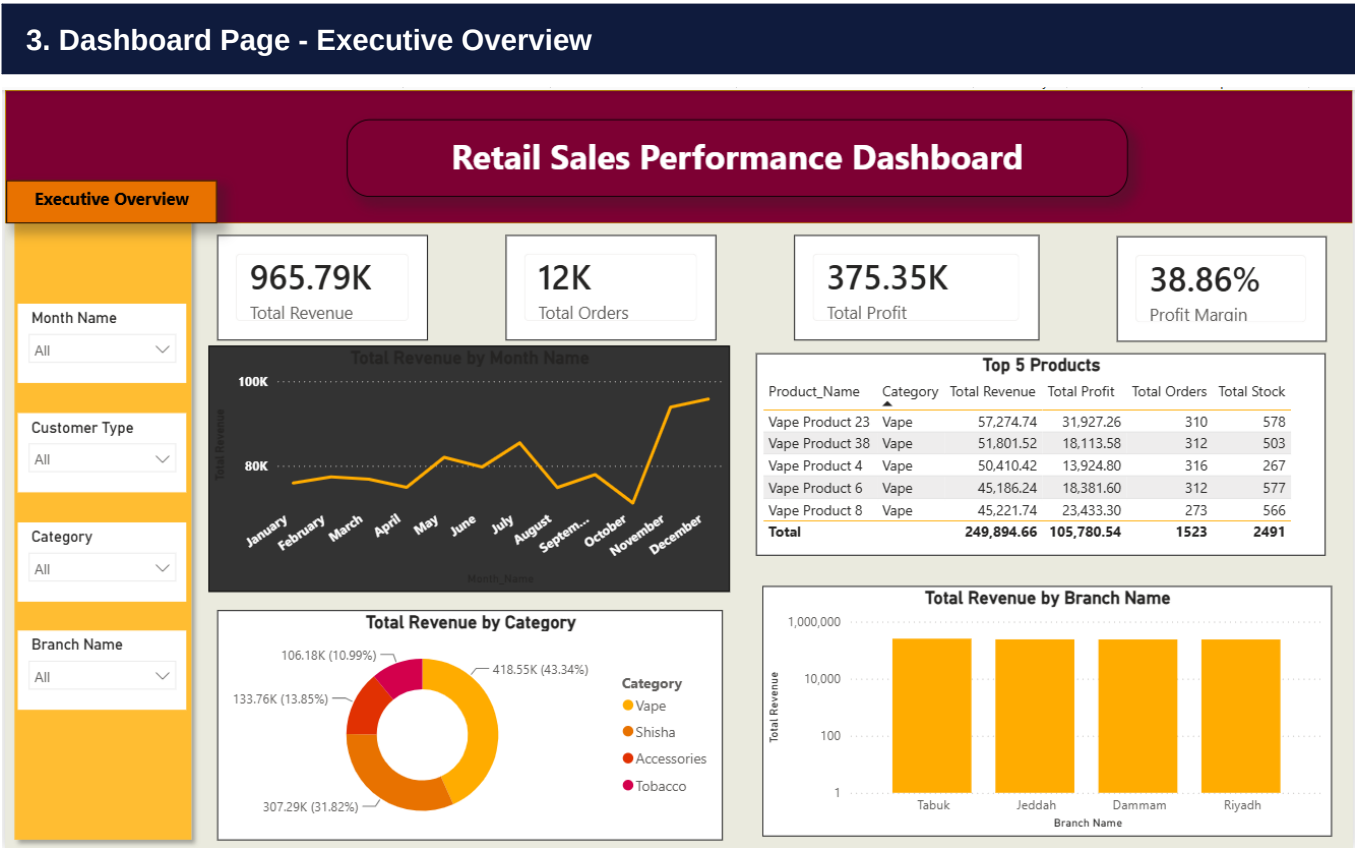
Area	Key Questions
Sales Performance	Which branches and product categories generate the highest revenue? What are the main revenue trends over time? Which products drive the highest sales contribution?
Customer Behavior	How many customers are new vs returning? What share of revenue comes from returning customers? Which customers or segments provide higher value?
Inventory & Operations	Which products are at risk due to low stock? Which branches have the highest inventory pressure? How many products reached reorder threshold?
Management KPIs	What are the core KPIs management should track regularly? Which areas require operational action?

Tools & Technologies

- Power BI for dashboard development and interactive reporting.
- DAX for business calculations and KPI measures.
- Power Query for data transformation and preparation.
- Data modeling for relationships, metric consistency, and reporting structure.
- Excel/CSV datasets as the operational data source format.

Measurement Approach

The dashboard was organized around executive KPIs, sales trends, customer groups, branch performance, product contribution, and inventory risk. The report translates dashboard outputs into business insights and recommendations that can be used by managers for regular performance reviews.



Key Metrics



Observations

- The dashboard provides a consolidated view of revenue, orders, profit, and profit margin.
- Revenue increases strongly toward Q4, especially in November and December.
- Vape products represent the largest category contribution at approximately 43.34% of revenue.
- Top 5 product performance is available for fast management review of product contribution and stock position.

4. Dashboard Page - Sales Performance Analysis

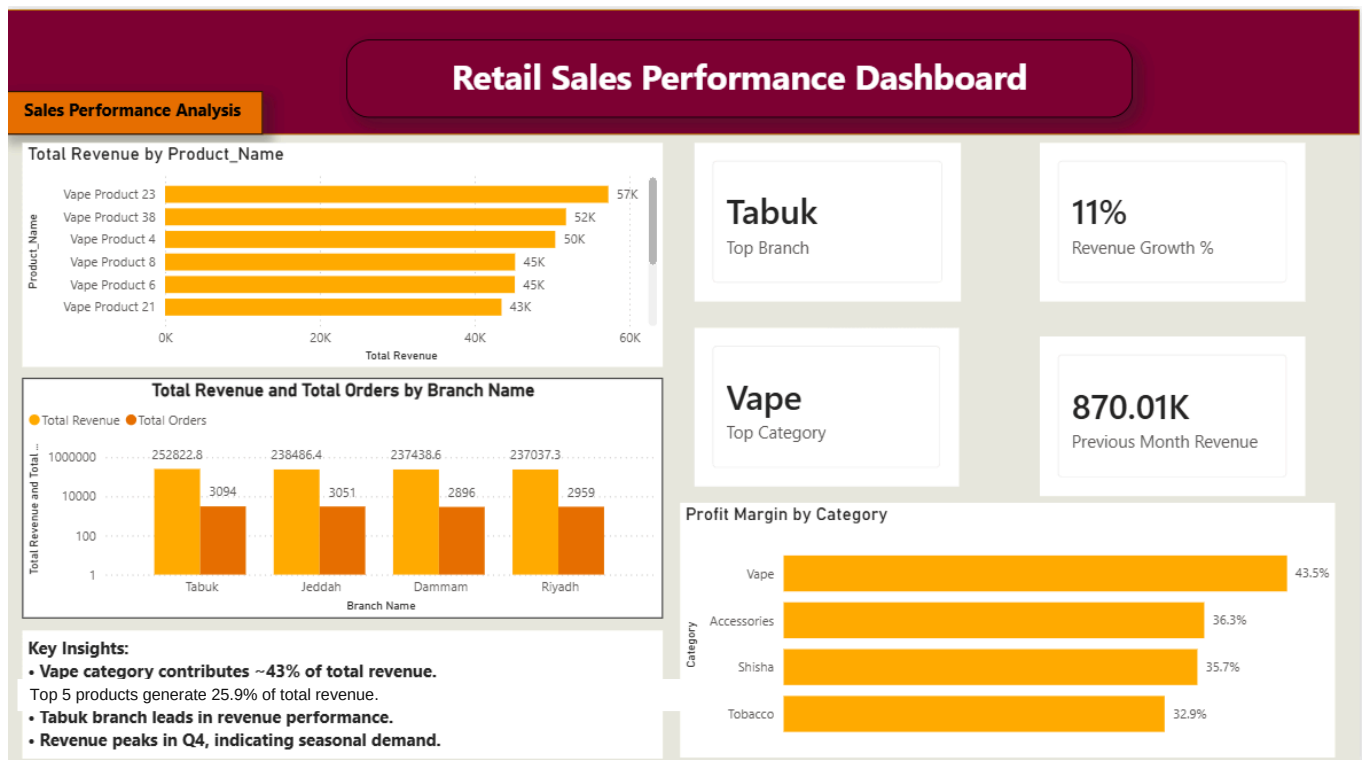


Figure 2. Sales Performance Analysis dashboard showing product revenue, branch revenue/order comparison, category profitability, top branch, top category, revenue growth, and previous month revenue.

Key Findings

- Tabuk is identified as the top branch by revenue performance.
- The Vape category is the top category and contributes roughly 43% of total revenue.
- The top five products generate approximately 25.9% of total revenue.
- Revenue growth is 11%, with previous month revenue at 870.01K.
- Profit margin is strongest in the Vape category at 43.5%, followed by Accessories at 36.3%.

Business Interpretation

The sales dashboard highlights the products, categories, and branches that drive revenue. This helps management prioritize inventory availability for high-performing products and understand where sales performance is strongest across branches.

5. Dashboard Page - Customer Behavior Analysis

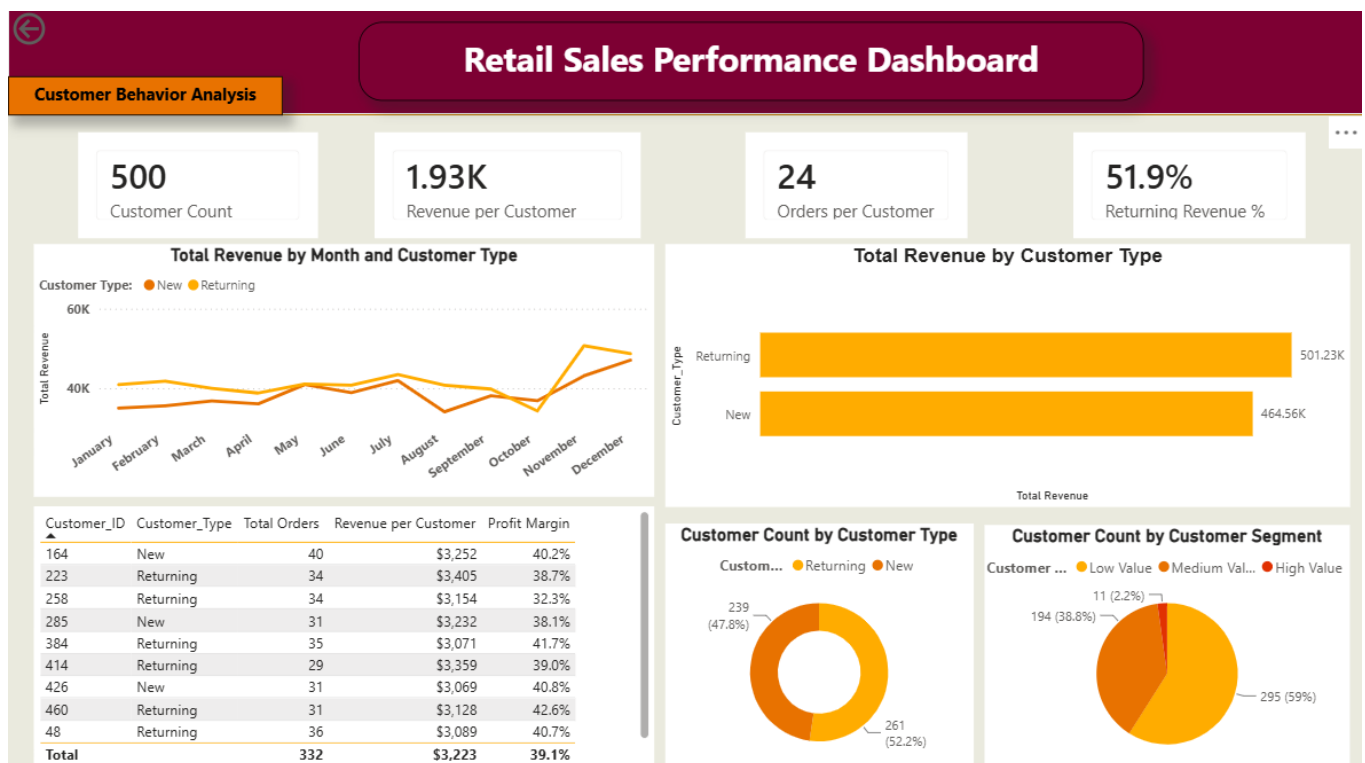


Figure 3. Customer Behavior Analysis dashboard showing customer count, revenue per customer, orders per customer, returning revenue percentage, customer type trends, and customer segmentation.

Key Findings

- The dashboard tracks 500 customers with revenue per customer of 1.93K.
- Orders per customer is 24, showing strong repeat purchase activity.
- Returning customers contribute 51.9% of total revenue, slightly higher than new customers.
- Customer count is balanced between new customers (52.2%) and returning customers (47.8%).
- Most customers fall into the low-value segment, while a small high-value group represents a priority retention opportunity.

Business Interpretation

The customer analysis supports retention planning by separating new and returning customers and identifying customer value groups. This enables targeted campaigns, loyalty initiatives, and stronger customer relationship management.

6. Dashboard Page - Inventory & Operations Analysis

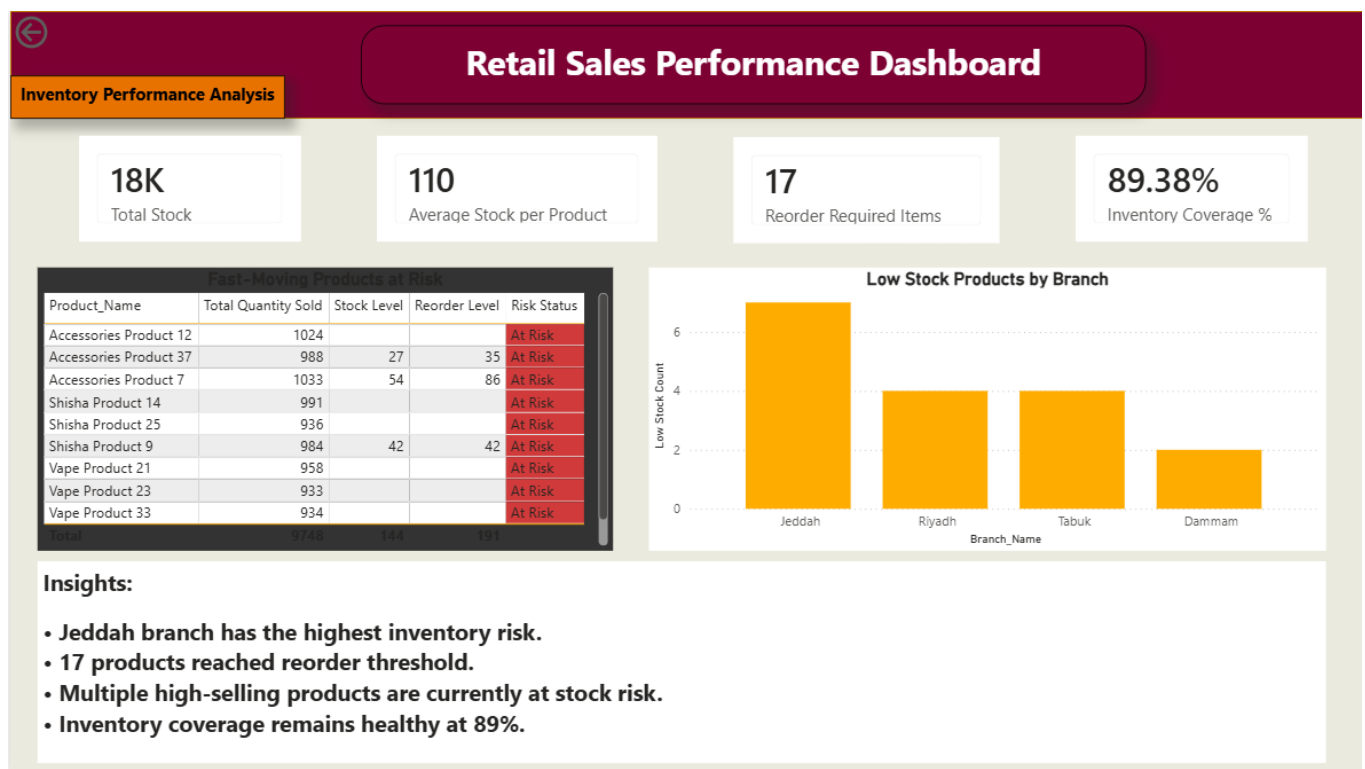


Figure 4. Inventory & Operations Analysis dashboard showing total stock, average stock per product, reorder-required items, inventory coverage, fast-moving products at risk, and low-stock products by branch.

Key Findings

- Total stock is approximately 18K, with average stock per product of 110.
- 17 products reached reorder threshold and require management attention.
- Inventory coverage remains healthy at 89.38%, but fast-moving items still show risk exposure.
- Jeddah branch has the highest inventory risk based on low-stock product count.
- Several high-selling products are flagged as at risk, which may affect future sales if not replenished quickly.

Business Interpretation

The inventory dashboard connects product demand with stock availability. This helps the business prevent stockouts for fast-moving items, monitor branch inventory pressure, and improve replenishment planning.

7. Recommendations and Project Outcome

Business Recommendations

- Improve replenishment planning for high-demand and fast-moving products.
- Monitor the 17 reorder-required products through a regular weekly inventory review.
- Prioritize Jeddah branch for inventory risk follow-up due to the highest low-stock exposure.
- Use customer segmentation to target medium- and high-value customers with retention campaigns.
- Track returning customer revenue as a customer loyalty and retention indicator.
- Use branch-level performance to identify operational differences and improve stock allocation.
- Expand executive KPI monitoring to support regular business review meetings.

Project Outcome

The Power BI dashboard provided a centralized reporting solution for retail management. It combined sales performance, customer behavior, inventory risk, and branch operations into a single BI environment, allowing faster and more structured decision-making.

The project demonstrates practical capability in business intelligence reporting, KPI design, data modeling, DAX measures, dashboard design, data storytelling, and operational analytics.

Files Included

File	Description
Saudi_Retail_BI_Dashboard.pbix	Power BI dashboard file.
Saudi_Retail_BI_Dashboard_Report.pdf	Business intelligence report documenting the project overview, questions, dashboards, insights, and recommendations.
README.md	GitHub project documentation.

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Note: This report is suitable for GitHub and portfolio presentation. Client/store identity and sensitive data are intentionally anonymized.